

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	SWTID-2026-4638
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	AI Caption Generator	Predicts whether a credit card application will be approved or rejected	Random Forest Model	Done	High
2	Data Preprocessing	Cleans and transforms applicant data before prediction	Data Processing Module	Done	High
3	Feature Encoding	Converts categorical inputs into machine-readable format	Encoding Module	Done	Medium
4	User Input Form	Allows users to enter applicant and financial details	Frontend UI	Done	Medium
5	Input Validation	Validates user inputs before processing	Flask Backend	Done	Medium
6	Risk Audit Report	Displays prediction result with applicant risk summary	Report Generator	Done	Medium
7	Prediction History Logging	Stores prediction details in JSON format for future reference	Logging Module	Done	Low
8	Responsive Web Interface	Provides a mobile-friendly and responsive user interface	HTML/CSS + Bootstrap	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	User Input Form, Responsive UI, Prediction Result Display
2	Backend / Logic	3	Data Preprocessing, Input Validation, Prediction Engine
3	Database / Storage	1	JSON Prediction History Logging
4	Machine Learning	1	Random Forest Credit Card Approval Prediction
5	Security / Authentication	1	Input Validation and Secure Request Handling